

Premium Kitchenware — Recognition × Recall Dissociation

AI Availability Score (AIAS) v0.17 Phase B Measurement

May 2026 · Pablo Ulpiano Gonzalez Castro, Third System™

Independent measurement for the AI mediation layer.

v0.17 brings the AIAS Presence Measurement Protocol v1.4 to the premium kitchenware substrate. The substantive pre-registered hypothesis FALSIFIES on panel inadequacy — worldwide $n = 10$, below the C1 floor of $n \approx 12$. The joint Identity-Load verdict resolves to AMBIGUOUS pending v0.18 indie fragrance. The substantive verdict is informationally weak; the secondary finding — the Recognition × Recall dissociation surfaced in the Japanese cell — is the most theoretically significant single result the AIAS programme has produced to date.

Three principal outcomes from a single Phase B measurement against the locked six-LLM reference panel (3 category queries × 6 LLMs = 18 cells per brand). **(1) H_Regime4_kitchenware FALSIFIED** on panel inadequacy — worldwide $n = 10$ below the pre-registered C1 floor of 12 after Phase A descope of Vermicular and Phase B exclusion of five brands. **(2) H_IdentityLoad_moderator (joint v0.16/v0.17) AMBIGUOUS** — the pre-registered joint verdict matrix at cell PARTIAL × FALSIFIED routes the Identity Load moderator hypothesis to v0.18 indie fragrance for resolution. **(3) Recognition**

× Recall dissociation — the Iwachu canonical case. Phase A C_p PASS at 6/6 anchoring (LLMs identify Iwachu as a Japanese cookware brand); Phase B EXCLUDED_E1a at 0/18 mention rate (LLMs do not surface Iwachu in unprompted premium-cookware category retrieval). The dissociation generalizes across the Japanese cell and motivates the v1.4 Methodology revision specifying AI Availability as a multi-component construct comprising Recognition and Recall.

AI assistants increasingly mediate brand discovery for consumer purchases. The AIAS™ Presence Measurement programme operationalises this with **AI Availability**: the conditions under which a brand surfaces in AI-generated category retrieval. v0.17 brings the v1.4 Protocol to the premium kitchenware substrate, the second leg of a joint test of the Identity Load moderator hypothesis (with v0.16 Kitchen Knives as the first leg, SSRN 6791999). The v0.17 brand panel comprises 16 brands stratified across three tradition cells: **european** (Le Creuset as primary pivot, Staub, Mauviel, Demeyere, Fissler, de Buyer; n = 6); **american** (All-Clad as primary pivot, Lodge, Made In, Field Company, Smithey, Hestan; n = 6); **japanese** (Vermicular as primary pivot, Iwachu, Sori Yanagi, Noda Horo; n = 4). Phase A and Phase B both completed against the locked six-slot reference panel.

H_Regime4_kitchenware FALSIFIED on panel inadequacy. Post-Phase-B worldwide n = 10, below the pre-registered C1 floor of n = 12. The Japanese cell collapsed entirely (0/3 eligible after Vermicular's Phase A descope and Iwachu/Sori Yanagi/Noda Horo's Phase B exclusions). The American cell sustained partial attrition (Field Company and Smithey EXCLUDED_E1a at 0/18 mentions). The European cell remained intact at 6/6 eligible. Per pre-reg §2 decision rule (→ C1 → C2): **FALSIFIED**. The pre-registration anticipated this outcome path explicitly; no post-hoc decision rule modification or threshold adjustment was applied.

H_IdentityLoad_moderator (joint v0.16/v0.17) AMBIGUOUS. The joint Identity-Load moderator hypothesis is tested across v0.16 Kitchen Knives (PARTIAL) and v0.17 Premium Kitchenware (FALSIFIED). The pre-registered joint verdict matrix at cell PARTIAL × FALSIFIED routes to **AMBIGUOUS pending v0.18** indie fragrance. v0.18 is the deciding test of the Identity Load moderator on a same-language English-only substrate, removing the cross-cultural confound that contaminated v0.17's Japanese-cell test.

Recognition × Recall dissociation — the methodological headline. Iwachu achieved Phase A C_p PASS at 6/6 (every LLM in the reference panel identifies Iwachu as a Japanese cookware brand when asked to disambiguate the canonical name) and Phase B EXCLUDED_E1a at 0/18 mentions (no LLM surfaces Iwachu in unprompted premium-cookware category retrieval across three queries and six models). The dissociation generalised across the Japanese cell. The Iwachu canonical case demonstrates that AI Availability is multi-component: a brand can have full recognition anchoring without any recall presence. The v1.4 Methodology paper revision formalises this as the Recognition + Recall decomposition of AIAS Presence. In the author's assessment this is the most theoretically significant single result the AIAS programme has produced to date — of greater long-term value than the substantive verdict itself.

What We Measured

v0.17 measures Phase A C_p anchoring (Recognition) and Phase B mention rate (Recall) for premium kitchenware brands across six reference LLMs: claude-opus-4-5, claude-sonnet-4-5, gpt-4o, gpt-4o-mini, gemini-2.5-flash, gemini-2.5-flash-lite. The reference panel is locked across all v0.17 measurement; provider model substitutions during the acquisition window are documented in DEVIATIONS Entries 2 and 3 (substitutions made before any pivot-validation data acquisition). Single measurement wave at May 2026.

Brand panel. v0.17 registry (`brands_kitchenware_v0.17.json`) covers three tradition cells: **european** (Le Creuset as primary pivot, Staub, Mauviel, Demeyere, Fissler, de Buyer; $n = 6$); **american** (All-Clad as primary pivot, Lodge, Made In, Field Company, Smithey, Hestan; $n = 6$); **japanese** (Vermicular as primary pivot, Iwachu, Sori Yanagi, Noda Horo; $n = 4$, with Noda Horo pre-registered as `borderline_classification: true`, `borderline_resolution_at: phase_b_topic_id`). The panel was pre-registered at worldwide $n = 16$ pre-floor with conservative anticipated attrition of 1–3 brands.

Phase A cascade. The European and American primary pivots achieved C_p PASS at full 6/6 anchoring without cascade. The Japanese primary pivot Vermicular C_p FAILED at 4/6: two reference models (gpt-4o-mini at slot 4; gemini-2.5-flash at slot 5) led their responses with non-substrate referents ('multiple meanings' framing and adjective definition respectively). The cascade activated Japanese cell ordinal 2 — Iwachu — which achieved C_p PASS at 6/6 with classifier rationales reading uniformly as variations on 'Japanese manufacturer of cast iron cookware.' Iwachu's Iwate nambu-tekki heritage anchors cleanly. Phase A formally closed at git tag **v0.17-phase-a-locked** (commit **8cdf0cd**) with a 15-brand operational panel.

Phase B LLM-substrate measurement. v0.17 implements the v1.4-canonical LLM-substrate Phase B specification (provisional in v0.17 per DEVIATIONS Entry 7, canonical in v1.4 Methodology paper). Three tradition-agnostic category queries: *What are the best premium cookware brands?*; *Recommend high-quality cookware brands for serious home cooks.*; *What cookware brands do professional chefs use?* Each query issued once per reference slot — 18 measurement cells per brand (3 queries $\times$ 6 LLMs). Mentions detected via mechanical alias generation: canonical and lowercase variants of the brand name, plus hyphen-substitution and space-substitution variants. The Noda Horo borderline classification resolved at Phase B as OUT_OF_SCOPE (enamelware-only; not in-scope premium cookware) per the protocol-mandated `borderline_resolution_at` schedule.

Verdict and decision rules. All pre-registered conditions and thresholds locked at git tag **v0.17-prereg-r1** (commit **3ebe426**, 19 May 2026 UTC) prior to any acquisition. Phase B locked at git tag **v0.17-phase-b-locked** (commit **54c83ec**). Verdicts follow mechanically from the pre-registered decision rules. No post-acquisition rule modification, threshold adjustment, or interpretive reframing has been applied. Methodological findings in Pattern 3 are post-hoc observations that motivated the v1.4 Methodology paper revision but did not affect the substantive verdict on `H_Regime4_kitchenware`.

FINDING 01

Phase B mention-rate distribution — European saturates, Japanese collapses

Premium kitchenware — Phase B mention rates across the 15-brand panel

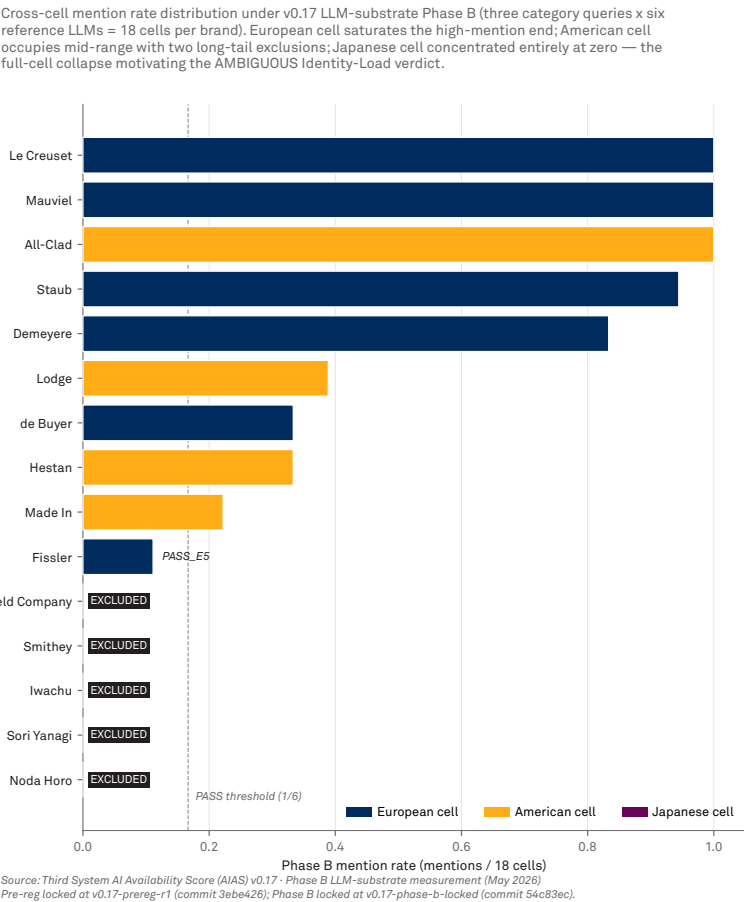


Figure 1 · Phase B mention-rate distribution across the v0.17 operational panel. Each bar is a brand; mention rate is the proportion of 3 (category queries) × 6 (reference LLMs) = 18 measurement cells in which the brand was mentioned. Sorted descending; colored by tradition cell. The European cell saturates the high-mention end (Le Creuset, Mauviel, Staub at or near 1.0). The American cell occupies the mid-range (Lodge, de Buyer, Hestan in the 0.22–0.39 band), with two long-tail brands (Field Company, Smithey) excluded at zero mentions. The Japanese cell sits entirely at zero — the full-cell collapse that motivates the AMBIGUOUS Identity-Load verdict and the v0.18 forward action.

The cross-cell distribution sets the empirical stage. Each of the 15 brands in the post-Phase-A operational panel was queried against the six-LLM reference panel under three tradition-agnostic category queries, producing 18 measurement cells per brand. The mention rate is the proportion of those 18 cells in which the brand was mentioned by the LLM in its response. The distribution is steeply asymmetric across tradition cells.

European cell saturates the high-mention end. Le Creuset, Mauviel, and All-Clad (American) reach 18/18 = 1.000 — mentioned in every cell. Staub at 17/18, Demeyere at 15/18, Lodge at 7/18, de Buyer and Hestan at 6/18, Made In at 4/18. Fissler sits at 2/18 — below the 1/6 PASS threshold but above the EXCLUDED_E1a floor, classified PASS_E5 per the v1.4 tier vocabulary.

Long-tail American attrition. Field Company and Smithey — both small-batch American specialty makers — sit at 0/18 mentions across all 18 cells. EXCLUDED_E1a. Both brands have stable identity in English-language commerce but do not surface in LLM-generated category retrieval at any frequency. This is the structural pattern v1.4 §5.2 documents as long-tail LLM-substrate attrition: the LLM panel concentrates its category responses on the most prominent five-to-ten brands and rarely mentions beyond that band.

Japanese cell collapses entirely. Iwachu, Sori Yanagi, and Noda Horo all sit at 0/18 — zero mentions across the entire measurement matrix. The full-cell collapse is hard to disambiguate from Western-LLM training-data bias on a cross-cultural substrate (see Limitations and Pattern 3). The pre-registration anticipated panel-collapse outcomes as a possible path; the pre-reg §3.2 worst-case scenario was full Japanese collapse leaving worldwide $n = 12$ at the C1 floor exactly. The observed outcome combined full Japanese collapse with partial American attrition, producing $n = 10$ — two brands below the C1 boundary.

FINDING 02

H_Regime4_kitchenware FALSIFIED on panel inadequacy — the C1 floor breach

Panel attrition by tradition cell — pre-Phase-A versus post-Phase-B
Worldwide n dropped from 16 to 10 across two attrition stages, breaching the C1 adequacy floor of 12 by a deficit of 2. The European cell remained intact; the American cell lost two long-tail brands; the Japanese cell collapsed entirely (Vermicular at Phase A; Iwachu, Sori Yanagi, Noda Horo at Phase B). This pattern produces the FALSIFIED-on-panel-inadequacy verdict for H_Regime4_kitchenware.

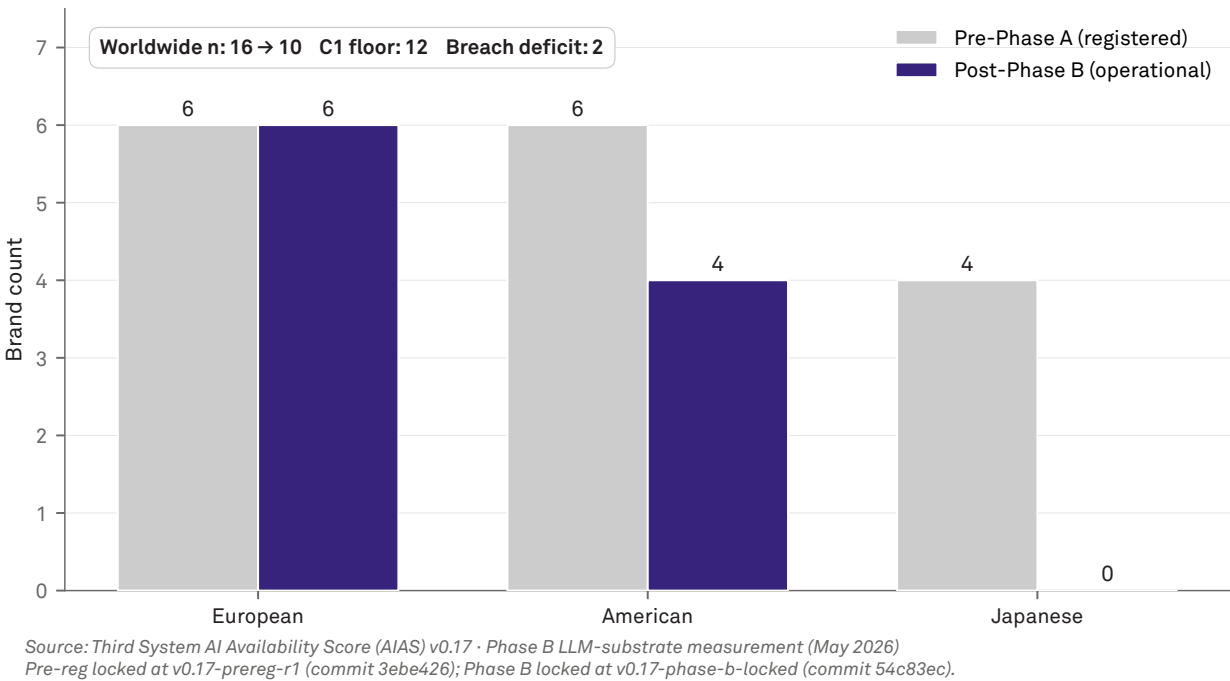


Figure 2 · Per-cell brand survival from pre-registered panel (pre-Phase A) to operational panel (post-Phase B). Worldwide n dropped from 16 to 10 across two attrition stages — Vermicular at Phase A (C_P FAIL at 4/6); Field Company, Smithey, Iwachu, Sori Yanagi, Noda Horo at Phase B. The deficit is 2 below the pre-registered C1 floor of 12. The European cell remained intact at 6/6; the American cell lost two long-tail brands; the Japanese cell collapsed entirely. This pattern produces the FALSIFIED-on-panel-inadequacy verdict for H_Regime4_kitchenware per the pre-reg decision rule $\neg C1 \vee \neg C2$.

The substantive pre-registered hypothesis.
H_Regime4_kitchenware tests whether the premium kitchenware substrate exhibits the Regime 4 pattern of AI-mediated retrieval per the v1.2 four-regime taxonomy, conditional on panel adequacy and effect-size persistence under tradition and age controls. Three pre-registered conditions: **C1** (n ≥ 12 eligible brands at worldwide); **C2** (Spearman ρ satisfies Regime 4 boundary); **C3** (partial ρ controlling for age and tradition retains the boundary).

Decision rule: **FALSIFIED** if $\neg C1 \vee \neg C2$.

C1 is breached. Worldwide eligible n = 10 at end of Phase B. C1 floor is n ≥ 12. Deficit: 2 brands below the boundary. C2 and C3 are not evaluated because the verdict resolves at C1. Per the pre-reg §2 decision rule: **FALSIFIED on panel inadequacy**. The pre-registration anticipated this outcome path explicitly — §3.2 names full Japanese cell collapse as the worst-case scenario the panel design accommodates.

Two attrition stages. The worldwide n dropped from the pre-registered 16 to the operational 10 across two stages. Stage 1 — Phase A: Vermicular C_p FAILED at 4/6 and was descope, dropping the Japanese cell from 4 to 3. Stage 2 — Phase B: Field Company and Smithey EXCLUDED_E1a from the American cell (American 6 → 4); Iwachu, Sori Yanagi, Noda Horo EXCLUDED_E1a from the Japanese cell (Japanese 3 → 0). The European cell remained intact at 6/6 throughout.

Joint Identity-Load verdict: AMBIGUOUS. The pre-registered joint verdict matrix combines v0.16 (PARTIAL on kitchen knives) with v0.17 (FALSIFIED on kitchenware) to resolve the Identity Load moderator hypothesis. At cell PARTIAL × FALSIFIED, the matrix routes to **AMBIGUOUS pending v0.18**

indie fragrance. v0.18 will test the moderator on a same-language English-only substrate, removing the cross-cultural confound and providing the deciding evidence.

Limitation acknowledged. The substrate-substitution from Trends to LLM-substrate Phase B (DEVIATIONS Entry 7) was applied after pre-registration lock. The Trends-substrate panel design anticipated 1–3 brands of attrition; the LLM-substrate produced 5 brands. The FALSIFIED verdict is mechanically correct against the pre-reg, but is partly a consequence of the substrate substitution rather than a clean test of the substantive hypothesis. v0.18 panel sizing will over-provision for LLM-substrate attrition at 30–50% per cell.

FINDING 03

Recognition × Recall dissociation — the Iwachu canonical case

Recognition × Recall dissociation — the Iwachu canonical case

The four pivot brands that received Phase A C_P measurement, plotted against Phase B mention rate. Le Creuset and All-Clad occupy the full-Presence quadrant; Vermicular failed Phase A (C_P at 4/6) and was descoped before Phase B; Iwachu sits alone in the recognition-only quadrant. The dissociation motivates the v1.4 Methodology revision specifying AI Availability as a multi-component construct.

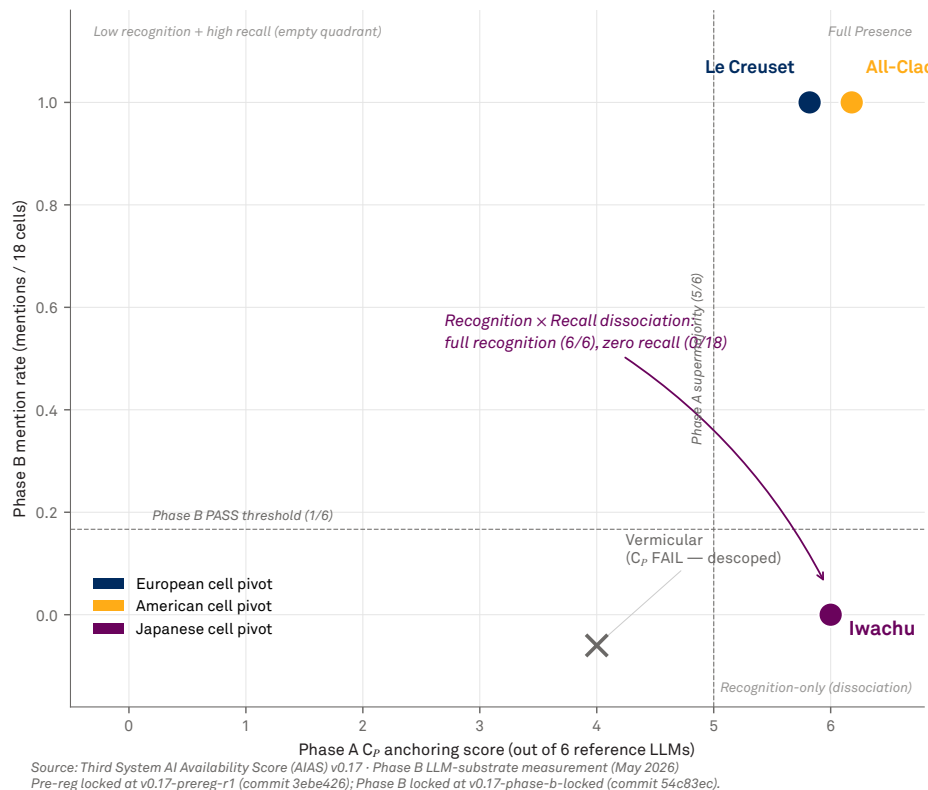


Figure 3 · Recognition × Recall dissociation. Phase A C_P anchoring score (x-axis, out of 6 reference LLMs) versus Phase B mention rate (y-axis, mentions / 18 cells) for the four pivot brands that received Phase A measurement. Le Creuset and All-Clad occupy the full-Presence quadrant (recognition and recall both saturated). Vermicular failed Phase A (C_P at 4/6) and was descoped before Phase B. Iwachu sits alone in the recognition-only quadrant: 6/6 Phase A anchoring (the LLM panel uniformly identifies Iwachu as a Japanese cookware brand) but 0/18 Phase B mention rate (no LLM surfaces Iwachu in unprompted premium-cookware category retrieval). The dissociation motivates the v1.4 Methodology revision specifying AI Availability as a multi-component construct comprising Recognition and Recall.

The methodologically significant finding. Iwachu — the Japanese cell pivot activated when Vermicular failed Phase A — produced a structurally robust dissociation between two measurement surfaces of LLM-mediated brand retrieval. At Phase A, the disambiguation query “Who or what is Iwachu?” produced six responses, each of which led with an identification statement placing Iwachu within the cookware substrate (LLM rationales read uniformly as “Japanese

manufacturer of cast iron cookware”). At Phase B, the same six reference LLMs queried with three unprompted category queries about premium cookware brands produced 18 response cells in which Iwachu was mentioned zero times. The brand has stable identity in the LLM’s representation but no recall presence in category-conditioned retrieval.

The dissociation is within-LLM. The same individual model that produced “Iwachu is a Japanese cookware brand” in response to the Phase A query did not include Iwachu in its response to the Phase B query “What are the best premium cookware brands?” Six LLMs, three queries, zero mentions — while Phase A produced 6/6 anchoring across the same six LLMs. This is not classifier disagreement, prompt artefact, or measurement noise. The model knows what Iwachu is. It does not surface Iwachu when asked about its category.

The pattern generalises across the Japanese cell. Sori Yanagi and Noda Horo, like Iwachu, received 0/18 mentions at Phase B. The full Japanese cell at Phase B exhibits the same dissociation pattern: stable identity (for Iwachu, confirmed at Phase A; for Sori Yanagi and Noda Horo, assumed by symmetry pending direct measurement) but zero recall presence in unprompted category retrieval.

AI Availability is multi-component. The Iwachu canonical case is the empirical anchor for the v1.4 Methodology revision specifying AI Availability as a multi-component construct with

Recognition and **Recall** as the two principal components. The substantive theoretical interest is in the Recall component: consumer behaviour in AI-mediated commerce depends on what AI intermediaries spontaneously surface (Recall), not on what they can identify when prompted (Recognition). A brand can have full Recognition with zero Recall; such a brand has partial AI Availability but is functionally invisible in AI-mediated category retrieval. The v1.4 Methodology paper formalises Recognition + Recall as the canonical decomposition of AIAS Presence and reweights the Presence composite accordingly.

This is the most theoretically significant single result the AIAS programme has produced to date. The implication for the broader Tri-System theoretical framework (manuscript in development for Routledge Studies in Marketing): AI Availability — the third system of brand availability alongside Mental Availability and Physical Availability — requires multi-component measurement. A single-metric proxy for AI Availability is inadequate. The Iwachu canonical case is the empirical demonstration.

Limitations

Panel design pre-substrate-substitution. The v0.17 pre-registration panel was sized for Trends-substrate Phase B attrition (1–3 brands anticipated). The substrate substitution from Trends to LLM-substrate Phase B (DEVIATIONS Entry 7) was implemented after pre-registration lock and produced 5 brands of attrition — approximately twice the anticipated rate. The pre-registered FALSIFIED verdict is mechanically correct against the pre-reg, but is partly a consequence of the substrate substitution rather than a clean test of the substantive hypothesis. v0.18 panel sizing will over-provision for LLM-substrate attrition.

Cross-cultural confound on Japanese cell. The Japanese cell’s full collapse at Phase B is hard to disambiguate from Western-language LLM training-data bias on a cross-cultural substrate. Iwachu (Iwate nambu-tekki heritage), Sori Yanagi (designer-craft heritage), and Noda Horo (enamelware) all carry high Identity Load. By the substantive Identity-Load prediction, these brands should have shown stronger AI Availability than equivalently-positioned but lower-IL brands. The opposite pattern was observed. The v0.17 data does not distinguish between Reading 1 (Identity Load does not moderate AI Availability) and Reading 2 (Western-language training-data bias is large enough on cross-cultural substrates to swamp the Identity Load signal). v0.18 indie fragrance, with entirely English-language category surface, is the immediate forward action to disambiguate.

Single substrate, single time window. v0.17 measures one substrate at one time window (May 2026). The Recognition × Recall dissociation finding generalises within v0.17 across cells and brands but has not yet been replicated across substrates or time windows. v0.18 will provide the cross-substrate replication; the time-window replication is a v1.5+ forward action.

Mechanical alias generation, not semantic. Phase B mention detection uses mechanical alias generation: canonical and lowercase variants, plus hyphen and space substitution. Semantic aliases (e.g., “Field and Company” for “Field Company,” or “Knife Made In” for “Made In”) are not generated. The v1.4 brand registry schema specifies an optional aliases field for brands requiring semantic-variant matching. None was registered for v0.17 — future programmes may include them.

Single classifier model at Phase A. Phase A C_p classification was automated via claude-opus-4-7 (DEVIATIONS Entry 4; provisional v1.4 §6.4.2). Inter-rater reliability with operator

judgment was not measured at v0.17; this is a v1.5+ forward action. The substrate-specific failure mode — the Vermicular “multiple meanings” cascade trigger — was empirically successful in surfacing the Iwachu activation. But the classifier’s decision boundary remains under-validated against operator judgment.

What's Next

v0.17 publication sequence. Brand-format report (this document), SSRN working paper, and OSF deposit ship together under git tag **v0.17-published**. The companion SSRN paper develops the substantive analysis alongside the formal hypothesis results and elevates the Recognition × Recall dissociation as the methodological headline.

v0.18 indie fragrance — the deciding test. The pre-registered joint verdict matrix routes the Identity-Load moderator hypothesis to v0.18 for resolution. Indie fragrance is entirely English-language, removing the cross-cultural confound that contaminated v0.17. The IL gradient within indie fragrance can be tested on equal LLM-coverage terms. v0.18 panel sizing will over-provision approximately 30–50% for LLM-substrate attrition per cell, per the v0.17 lesson.

AIAS Protocol v1.4 Methodology paper. Already shipped as SSRN forthcoming. v1.4 supersedes v1.2 and v1.3 and formalises Recognition + Recall as the canonical decomposition of AIAS Presence. The v0.17 program is the empirical anchor for the multi-component construct claim.

Tri-System Brand Growth and Routledge monograph. The Tri-System MSI Working Paper bibliography is updated to incorporate v0.17 and AIAS Protocol v1.4 as empirical anchors for the framework's AI Availability axis. The Recognition × Recall dissociation strengthens the framework's multi-component AI Availability claim. The Routledge monograph (*The Third System: AI Availability and the Architecture of Brand Growth*, in development) carries the finding forward as the empirical anchor for the multi-component construct of AI Availability.

AIAS 1.0 and Phase 4. The full AIAS composite (Presence, Ranking, Consistency, Coverage, Grounding, Sentiment) remains a multi-year arc. v0.17 advances the Presence component with the Recognition × Recall decomposition; Phase 4 will ship the remaining five components contingent on Phase 3 construct-validity results.

Hypothesis Scoring

Two pre-registered hypotheses for v0.17. All thresholds and decision rules locked at git tag v0.17-prereg-r1 (commit 3ebe426, 19 May 2026 UTC) prior to any acquisition. The substantive hypothesis H_Regime4_kitchenware tests Regime 4 replication on the premium kitchenware substrate; the joint hypothesis H_IdentityLoad_moderator tests Identity Load moderation across v0.16 (kitchen knives) and v0.17 (premium kitchenware) as the two medium-IL substrates. Methodological findings (Recognition × Recall dissociation) are post-hoc observations from v0.17 and are not shown in this scoring table; their formal status is detailed in the v1.4 Methodology paper.

H	Pre-registered prediction	Result	Status
H_Regime4_kitchenware (substantive, worldwide)	$n \neq 12$ AND $ bivariate\ p(AI,\ Trends) $ satisfies Regime 4 boundary AND partial $p(AI,\ Trends age,\ tradition)$ retains the boundary at t_1 and t_2	$n = 10$ (worldwide eligible after Phase A descope of Vermicular and Phase B exclusion of 5 brands); C2 and C3 not evaluated — verdict resolves at C1	FALSIFIED on panel inadequacy
H_IdentityLoad_moderator (joint v0.16 / v0.17)	Joint verdict matrix combining v0.16 (PARTIAL on kitchen knives) with v0.17 substantive outcome; routes to CONFIRMED / PARTIAL / AMBIGUOUS / FALSIFIED per pre-reg matrix	v0.16 PARTIAL × v0.17 FALSIFIED → matrix cell routes to AMBIGUOUS pending v0.18 indie fragrance for resolution	AMBIGUOUS

Hypothesis Details

Per-hypothesis claim, operationalisation, and result. Both hypotheses pre-registered at v0.17-prereg-r1 (commit 3ebe426, 19 May 2026 UTC). Decision-rule language and threshold values quoted verbatim from the pre-registration.

Substantive hypothesis (v0.17). Brand AI Availability on the premium kitchenware substrate is tested for the Regime 4 (Covariate-saturated weak) signature established on premium facial skincare (v0.11), personal finance apps (v0.13), premium tea (v0.14, v0.15), and kitchen knives (v0.16 PARTIAL). Three conditions tested: **(C1)** worldwide $n \neq 12$ brands surviving Phase A pivot validation and Phase B mentionability; **(C2)** Spearman p on the operational panel satisfies the Regime 4 boundary; **(C3)** partial p controlling for age and tradition retains the boundary. Decision rule per pre-reg §2: **FALSIFIED** if $\neg C1 \rightarrow C2$. Result: $n = 10$ (worldwide eligible after two-stage attrition); $\neg C1$ satisfied; FALSIFIED follows mechanically. C2 and C3 not evaluated. **FALSIFIED on panel inadequacy.**

Joint hypothesis. Identity Load moderates the strength of the Regime 4 pattern across substrates. v0.16 (kitchen knives, medium IL) and v0.17 (premium kitchenware, medium IL) form the two-leg test; v0.18 (indie fragrance, higher IL) and future low-IL substrates extend the test in either direction. The pre-registered joint verdict matrix specifies, for each combination of v0.16 and v0.17 outcomes, the corresponding joint Identity-Load verdict. The relevant cell for v0.17’s outcome path is PARTIAL × FALSIFIED — the matrix routes to **AMBIGUOUS pending v0.18**. v0.18 indie fragrance will be the deciding test on a same-language English-only substrate, removing the cross-cultural confound that contaminated v0.17’s Japanese cell.

About the Third System

The Third System describes a shift in how consumers make decisions: instead of browsing and comparing options directly, many now rely on AI systems to generate recommendations. In this environment, brands compete not only for human attention but for inclusion within AI-generated outputs. This creates a new competitive layer where visibility is determined by how AI systems interpret and surface brands.

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Methodology

This report presents findings from Third System's AI Availability research program. The AI Availability Score (AIAS) is an early-stage measurement framework introduced in Tri-System Brand Growth (Gonzalez Castro, 2026). Findings should be used directionally. Construct validity — the correlation between AI Availability and consumer behavior — has not yet been established and is the subject of Phase 3 of the research program. Methodology is published in full and versioned at thirdsystem.ai/methodology (current: v1.1). Registry construction is curated and reflects the framework's view of each category's competitive set; revisions are documented in the methodology log.

Citation

Gonzalez Castro, P. U. (2026). *Panel Inadequacy and the Recognition x Recall Dissociation on the Premium Kitchenware Substrate: AIAS v0.17*. Third System. thirdsystem.ai/v17-kitchenware-dissociation (SSRN forthcoming)

Underlying datasets

OSF project ec6wh, /v17/. Inputs: Phase A pivot validation across the six-LLM reference panel (claude-opus-4-5, claude-sonnet-4-5, gpt-4o, gpt-4o-mini, gemini-2.5-flash, gemini-2.5-flash-lite) with cascade activation of Iwachu after Vermicular C_p FAIL (DEVIATIONS Entry 6); Phase B LLM-substrate mention measurement (three tradition-agnostic category queries × six LLMs = 18 cells per brand); Phase B topic-ID resolution with Noda Horo borderline classification resolved as OUT_OF_SCOPE (enamelware-only); v0.17 registry (**brands_kitchenware_v0.17.json**, three tradition cells: european 6, american 6, japanese 4 pre-Phase A); classification ledger; Phase B response caches; scoring outputs; 3 chart PDFs from `build_charts_v17.py`; build scripts; this report; the matching SSRN working paper.

Companion SSRN working paper and AIAS programme cross-references. v0.17 working paper: *Panel Inadequacy and the Recognition x Recall Dissociation on the Premium Kitchenware Substrate: AIAS v0.17*. Cross-references: AI Availability foundational paper (SSRN 6659000); AIAS Presence Measurement Protocol v1.1 (SSRN 6722319); AIAS Presence Measurement Protocol v1.2 (SSRN 6761698); AIAS Presence Measurement Protocol v1.3 (SSRN 6797679); AIAS Presence Measurement Protocol v1.4 (SSRN forthcoming); v0.6 Cross-Category Findings (SSRN 6720959); v0.7 Phantom-Brand Persistence Phase 2 BBB (SSRN 6721779); v0.8 Discourse-Language Knives (SSRN 6728000); v0.9 Longitudinal Re-Baseline (SSRN 6736878); v0.10 Naive-Phantom Rate Stability (SSRN 6741163); v0.11 PM Software × Trends Construct Validity Pilot (SSRN 6745040); v0.12 Three Empirical Regimes (SSRN 6748341); v0.13 Four Empirical Regimes — Five-Category Construct-Validity Expansion (SSRN 6750498); v0.14 Kitchen Knives — Regime 4 Replication (SSRN 6755621); v0.15 Premium Tea Panel Expansion Robustness (SSRN 6768059); v0.16 Regime 4 Boundary and Discourse-Language Carryforward on Kitchen Knives (SSRN 6791999).

Methodology log: v0.17 follows AIAS Presence Measurement Protocol v1.4 (SSRN forthcoming). Pre-registration locked at git tag **v0.17-prereg-r1** (commit **3ebe426**) on 19 May 2026 UTC prior to Phase A acquisition. Phase A formally closed at git tag **v0.17-phase-a-locked** (commit **8cdf0cd**); Phase B formally closed at git tag **v0.17-phase-b-locked** (commit **54c83ec**). DEVIATIONS.md: eight entries documented contemporaneously; key entries are Entry 4 (operator judgement → automated LLM classifier provisional v1.4 §6.4.2); Entry 6 (Vermicular C_p FAIL → Iwachu cascade activation); Entry 7 (substrate substitution from Trends to LLM-substrate Phase B, provisional v1.4 §5.2). No pre-registered hypothesis, threshold, or routing rule was modified post-acquisition..

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